Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

Interest-rate surprises around Federal Reserve announcements are a clean measure of monetary policy shocks. But their estimated effects on unemployment and inflation since 1994 are often weak or have the wrong sign. We keep the surprise as it is and change only how its effects are traced through the economy. If the surprise is traced through the two-year yield alone, the wrong signs appear. If it is traced through financial conditions as a whole, measured relative to the neutral interest rate r^*, the effects have the expected signs and are precisely estimated. A policy move that raises the two-year yield 1 percentage point raises the unemployment rate about 0.4 percentage point and lowers industrial production about 1 percent within a year. It lowers core prices about 0.6 percent after two years. Including financial conditions matters most for the unemployment result. Measuring them relative to r^* matters most for the price result, which relies on data since 2020.

1 Introduction

How does monetary policy affect the economy? Standard theory says that a surprise tightening should raise unemployment and lower inflation. Yet empirical work often finds the opposite signs, or effects too imprecise to tell. The problem is worst in samples that start after the mid-1980s, when policy followed the economy more closely and left fewer clear shocks to learn from (Ramey, 2016).

Interest-rate surprises around Federal Reserve announcements were meant to solve this. The FOMC makes its decision before the announcement, so market moves in a narrow window around the announcement cannot have caused the decision. The change in interest-rate futures and Treasury yields in that window is then news about policy (Kuttner, 2001; Gürkaynak et al., 2005). These surprises measure the effects of policy on asset prices precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). But when they are used to measure effects on unemployment and inflation, the old puzzles often return.

Recent work offers three explanations, each with a fix.

- *Information.* An announcement reveals the Fed's view of the economy as well as its policy. A surprise tightening can then signal good news about the economy (Romer and Romer, 2000; Nakamura and Steinsson, 2018). Jarociński and Karadi (2020) separate the two using a simple idea: a pure policy tightening should lower stock prices, so when rates and stocks rise together, the surprise mostly carries information. The pure policy part gives a larger fall in prices (see also Miranda-Agrippino and Ricco, 2021).
- *Predictable surprises.* Part of each surprise can be predicted from public data released before the announcement, so it is not a clean shock. And after 1984 there are few large surprises to

learn from. Bauer and Swanson (2023a) add surprises from speeches by the Fed Chair and remove the predictable part. Their effects on the economy are larger and more precise (see also Bauer and Swanson, 2023b).

- *Timing.* Markets react within minutes, but the economist sees unemployment and inflation only as monthly averages. Jacobson et al. (2026) show that this mismatch alone can produce wrong signs. When they measure inflation daily, using online prices from 2008 to 2015 and keeping the same surprises and the same method, most of the wrong-signed response disappears.

The first two fixes change the surprise. The third changes how the outcome is measured. Gertler and Karadi (2015) and Caldara and Herbst (2019) add a fourth point: credit costs carry part of the effect, and leaving out credit spreads weakens it.

This note offers a new take. We keep the surprise as it is, and we keep the outcomes and the method fixed. We change only how we measure the variables that carry policy to the economy. Standard models carry the surprise forward through one interest rate, measured in nominal terms. That leaves out two basic parts of how monetary policy works.

First, policy affects spending through financial conditions as a whole: the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010). A surprise tightening moves long-term yields, stock prices and the dollar, not only the two-year yield. Policy analysts often summarize these conditions in a financial conditions index (Hatzius et al., 2010; Ajello et al., 2023).

Second, what matters for spending and inflation is the level of real interest rates compared with the neutral rate r^*. The neutral rate is the rate that neither speeds up nor slows down the economy. It fell a lot from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Kiley, 2020), and it shaped the long-run trend in yields (Bauer and Rudebusch, 2020). Since 1994, interest rates have trended down along with r^*, while activity has trended up. A model that links the level of rates to activity then partly picks up these opposite trends. It finds that higher rates go with stronger activity, which looks like a wrong-signed effect of policy.

The two parts need not matter equally for every outcome. Activity responds to changes in financial conditions. Inflation depends on how long real rates stay above or below r^*, through the slack this creates. So measuring rates relative to r^* should matter most for prices.

Our design is simple. The shock is the policy-news surprise of Nakamura and Steinsson (2018), measured in narrow windows around 252 FOMC announcements from 1994 to 2026. We read its effect on the two-year yield, and on a market-based financial conditions index (FCI), from these windows. We assume that unemployment and inflation do not move within the windows. A daily model carries the two-year yield and the FCI forward from the announcement day. Monthly unemployment and core inflation then respond to the monthly averages of these daily series, as in a standard monthly model (a vector autoregression, or VAR). Like Jacobson et al. (2026), we take the timing of each announcement within its month into account. The FCI and the order of steps—nominal, real, then relative to r^*—come from a companion note on financial conditions (FCI note, 2026). We fixed them before this analysis.

The evidence builds in four steps. Every step includes a smooth time trend (a quadratic) in the monthly equations.

- With the nominal two-year yield as the only channel, the post-1994 puzzles appear. The unemployment response changes with the number of past months in the model (lags). With 12 lags it is negative, and its 90 percent interval excludes zero. Below, "significant" means that the 90 percent interval excludes zero. Prices barely move, and industrial production rises.

- Adding the FCI makes the unemployment response standard, large and significant. But prices fall only a little, and industrial production does not fall.
- Measuring rates in real terms changes little.
- Measuring the two-year yield and the FCI relative to r^* completes the picture. Unemployment rises 0.36 to 0.40 percentage point after 12 months. The core price level falls 0.53 to 0.72 percent after 24 months. Industrial production falls about 1 percent after 12 months. All are significant for every number of lags.

Compared with the benchmark of Bauer and Swanson (2023a), our price effect is similar for two years and larger after that. Our effect on industrial production is somewhat smaller. Our effect on unemployment is about twice as large after one year and lasts longer.

The robustness checks show what drives these results. Financial conditions are essential for unemployment. Without the FCI, no adjustment of the two-year yield gives a reliable unemployment response. With the FCI, every adjustment does. Measuring conditions relative to r^* is essential for prices. The price effect is strong whenever the FCI is measured relative to the Laubach–Williams r^*, however the two-year yield is adjusted. The unemployment response holds up however we treat the pandemic, and it is larger in a sample that ends in 2019. The significant price effect, by contrast, depends mainly on data since 2020. The results are the same or slightly stronger with the cleaned surprises of Bauer and Swanson (2023a) and the pure-policy surprises of Jarociński and Karadi (2020). They also hold with a fuller method for computing uncertainty.

1.1 Connection to the literature

Our paper is closest in spirit to Jacobson et al. (2026). Like them, we hold the surprise and the method fixed and change one measurement. They change how often the outcome is measured. We change how the channel from policy to the economy is measured. They study inflation only; they do not estimate effects on activity, which is where our main result lies. Both changes remove wrong-signed responses, and both point to a mismatch between what the economist measures and what drives decisions: daily rather than monthly information in their case, financial conditions relative to r^* rather than a single nominal rate in ours. Our model also uses the timing of announcements within the month, which is central to their argument. Related work on timing includes Kilian (2024), Chudik and Georgiadis (2022), Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020).

Bauer and Swanson (2023a) and Jarociński and Karadi (2020) improve the surprise itself. In our design, a surprise enters only through its immediate effect on the two-year yield and the FCI, and through its timing. Using their cleaned surprises in our model changes the results little and makes them slightly stronger. So their fixes and ours work in the same direction and can be combined. Our approach also changes which assumptions the estimates rest on. It depends less on how strongly the surprise moves rates over a month, and less on its link to earlier news. But it relies on a different assumption: the economy responds to a given change in financial conditions in the same way, whatever caused the change. Gertler and Karadi (2015) and Caldara and Herbst (2019) are closest in emphasizing financial channels; we use a broad index and measure it relative to r^*, but we leave out credit spreads. The use of futures-based surprises in monthly models goes back to Faust et al. (2004), and the choice of surprise series can matter for the results (Bu et al., 2021; Brennan et al., 2025). To our knowledge, no earlier study traces the effects of these surprises through a broad financial conditions index measured relative to r^*.

2 Data and method

The surprise. The shock is the Nakamura–Steinsson policy-news surprise. It comes from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). It summarizes the changes in five short-term interest-rate futures around 252 FOMC announcements from February 1994 to September 2026. The window runs from 10 minutes before the statement to 60 minutes after the press conference starts. When there is no press conference, it is the 30-minute window around the statement. We scale the surprise so that the two-year Treasury yield rises 1 percentage point in the window. We measure the FCI's move in the same windows from the 10-year yield, the S&P 500 and the dollar, using the FCI's weights. The FCI rises 0.99 unit per percentage point rise in the two-year yield. It is close to one by design, because the weights are scaled so that one unit of the index equals a policy move that raises the two-year yield 1 percentage point. Unemployment and prices are not observed at that frequency, so we assume they do not move in the window.

Financial conditions. The FCI is the market index of FCI note (2026). It is a weighted sum of the 10-year Treasury yield, stock valuations (the cyclically adjusted price-earnings ratio, CAPE) and the real broad dollar. The weights come from how each of these moves with the same policy news. There are three versions, built in the same steps as in that note:

- *Nominal*: the series as measured.
- *Real*: the 10-year yield minus 10-year expected inflation.
- r^**-gap* (baseline): the real 10-year yield minus the Laubach–Williams r^*, and stocks measured by the gap between their earnings yield and r^*. We use the New York Fed's current estimates of r^*, based on Holston et al. (2023). These "one-sided" estimates use only data up to each date, with the model's parameters estimated on the full sample.

The two-year yield is adjusted the same way. It is nominal, then minus two-year expected inflation (from the Cleveland Fed; Haubrich et al., 2012), then also minus r^*. The daily series follow the daily 10-year yield, CAPE and dollar. We shift them smoothly within each month so that their monthly averages equal the monthly indexes exactly. Expected inflation and r^* move slowly, so the moves in the window are the same in every version.

The model. The model has a daily part and a monthly part. In the daily part, the two-year yield and the FCI (when included) depend on their own averages over the past 1, 5, 22 and 66 days. They also respond to this month's and last month's unemployment and inflation. We estimate this part over 1990–2026. In the monthly part, the unemployment gap and core PCE inflation (the price index for personal consumption expenditures excluding food and energy) depend on their own values in the past L months and on the past L monthly averages of the daily series. The unemployment gap is the unemployment rate minus the CBO estimate of its natural rate. The CBO estimate does not respond to the surprise, so the response of the gap equals the response of the unemployment rate. We estimate this part by ordinary least squares from February 1994 to September 2026, leaving out March–December 2020. We use 3, 6 or 12 monthly lags.

We assume that within each month, unemployment and inflation are set first, and markets then respond to them. In the month of the announcement, unemployment and inflation have not yet moved. To trace the effect of a surprise, we place its window effect on the announcement day. That day is drawn from the actual pattern of FOMC dates within the month. We then run the daily model forward, take monthly averages, and run the monthly model forward. This is a mixed-frequency VAR (Ghysels, 2016) that gives each day in a month the same weight, as a standard monthly VAR does. In large samples, this kind of VAR estimates the same effects as the alternative method of local projections, which regresses each outcome directly on the shock at

each future month (Plagborg-Møller and Wolf, 2021). A test does not reject equal daily weights for unemployment. It does reject them for inflation, but the gain in fit from unequal weights is very small.

The time trend. Every monthly equation includes a smooth quadratic time trend. Over 1994–2026, unemployment, inflation and interest rates contain slow movements that a short monthly model cannot capture. Rates drifted down along with r^*. Inflation was low in the 2010s and rose after 2020. The natural rate of unemployment may have shifted in ways the CBO estimate misses. Without a trend, these slow movements show up in the estimated effect of the financial variables, and trend gets mixed with transmission. A quadratic is the simplest smooth trend that allows inflation to fall and then rise. Because the trend is an assumption, Appendix Table A3 shows all the main steps without it. The nominal and real steps depend much more on the trend than the r^* step does. This suggests that measuring rates relative to r^* does much of the work of the trend.

The key assumption. We assume the surprise affects unemployment and prices only through the variables in the daily model. We also assume the economy responds to a given change in those variables the same way, whatever caused it. The first part is easier to accept when the financial variables are broad. The second part is the assumption behind any use of a financial conditions index to measure the stance of policy. It is a strong assumption. For example, stock prices partly anticipate future activity. We do not split the FCI into its parts, or into policy-driven and other movements. The index is meant to summarize transmission as a whole, and its weights already measure how each part responds to policy news.

We also tried a method that does not need this assumption: a direct regression of monthly unemployment on the daily surprises (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026). It gives wrong-signed estimates, driven largely by the 2007–09 announcements. Its intervals are also more than ten times as wide as ours, so it cannot tell our estimates apart from zero.

Uncertainty. To measure uncertainty, we use a bootstrap. In each of 200 draws, we redraw the announcements and the errors of the monthly equations at random, with replacement. Resampling the announcements also re-estimates the window effects and the timing of announcements. Section 4 reports a fuller version that also re-estimates the daily model and rebuilds the monthly data from the model. The FCI weights and r^* are held fixed throughout. We report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016). We also report 68 percent intervals, which Jarociński and Karadi (2020) report alongside 90 percent ones.

3 Results

Table 1 shows how the unemployment gap and the core PCE price level respond after 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 1: Responses to a policy move that raises the two-year yield 1 percentage point

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
	included	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	none	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
	included	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	included	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
B. Core PCE price level (percent)										
Nominal	none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
	included	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
Real	none	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
	included	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	included	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

Columns: months after the policy move; within each, the number of monthly lags in the model (3, 6 or 12). FCI "included": the FCI of the same step. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Steps are described in Section 2. All versions include a quadratic time trend. Bauer and Swanson (2023a), benchmark (read from their Figure 8): unemployment +0.19, +0.17, +0.15 at 12, 24, 36 months (their 90 percent band excludes zero only at 12 months); CPI (not core PCE) −0.36, −0.48, −0.6 (their bands exclude zero).

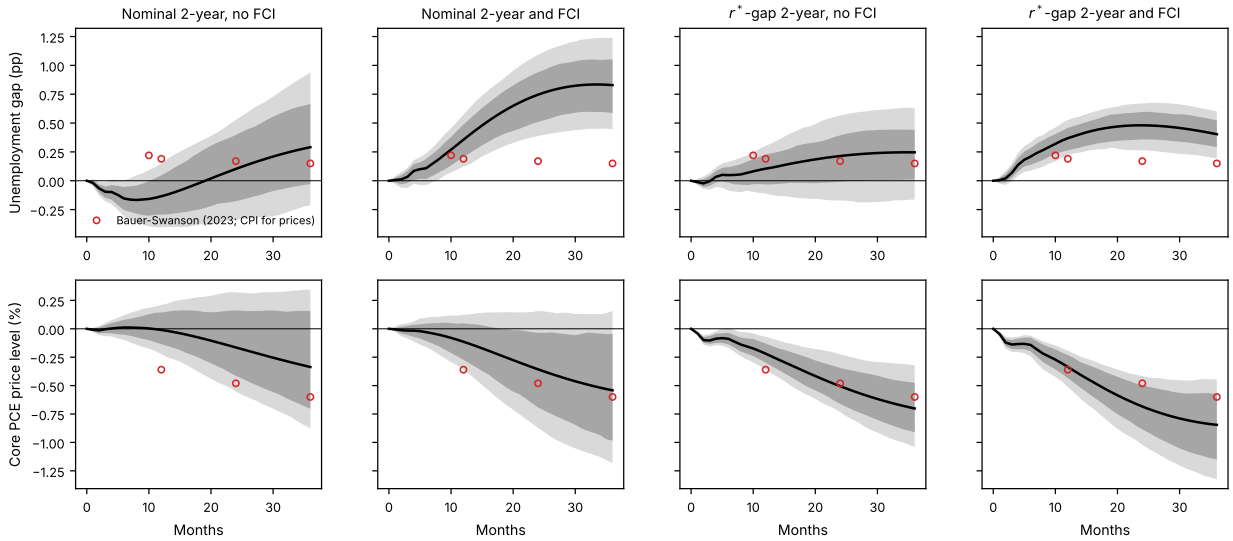


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent interval. Light band: 90 percent interval. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard approach reproduces the puzzles. Here the nominal two-year yield is the only channel. This is the policy measure that Bauer and Swanson (2023a) use. The unemployment response depends on the number of lags. With 12 lags, it is significantly negative (−0.27 after 12 months), so unemployment falls after a tightening. The price level barely moves.

This is the post-1994 pattern documented by Ramey (2016), and the one Bauer and Swanson (2023a) find with their unadjusted surprises. Industrial production rises significantly (Appendix Table A1).

Step 2: add financial conditions. Adding the FCI makes the unemployment response standard, large and significant at the 90 percent level at 12, 24 and 36 months and for every number of lags. Unemployment rises 0.28 to 0.44 percentage point after 12 months and 0.6 to 0.8 after 24 to 36 months. The price level falls 0.2 to 0.5 percent after 24 to 36 months. This fall is significant only at the 68 percent level, and only with 6 and 12 lags. Industrial production does not fall.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little. The companion note finds the same: after 1990, its nominal and real indexes are close. The price fall becomes significant at the 68 percent level at 12, 24 and 36 months and for every number of lags.

Step 4: rates relative to r^*. Next we measure the two-year yield and the FCI relative to r^*. Unemployment rises 0.36 to 0.40 percentage point after 12 months, 0.47 to 0.50 after 24 months and 0.35 to 0.43 after 36 months. These numbers are nearly the same for every lag length. The core price level falls 0.33 to 0.34 percent after 12 months, 0.53 to 0.72 percent after 24 months and 0.57 to 0.97 percent after 36 months. All of these are significant at the 90 percent level for every lag length. The unemployment response is smaller and fades sooner than with the nominal FCI. Without the FCI, the r^*-gap two-year yield gives a strong price effect. But the unemployment effect is then weak and depends on the number of lags (it is significant only with 3 lags).

Comparison with Bauer and Swanson (2023a). Their benchmark model covers 1973–2020. It uses surprises from FOMC announcements and Chair speeches, with the predictable part removed. Per percentage point on the two-year yield, it gives an unemployment response of about 0.19 percentage point after 12 months, 0.17 after 24 months and 0.15 after 36 months. It gives a CPI decline of about 0.36, 0.48 and 0.6 percent. Our step-4 price effects (for core PCE) are similar after 12 and 24 months and larger after 36 months (-0.57 to -0.97). Our unemployment effect is about twice as large after 12 months and 2.5 to 3 times as large after 24 to 36 months. Our intervals exclude zero through 36 months. Theirs include zero after about a year and a half. Their estimates use surprises from 1988 to 2019, Chair speeches and removal of the predictable part. Ours use FOMC announcement windows alone, with no such adjustment.

4 Robustness

How should the two-year yield be adjusted? Table 2 and Table 3 keep the FCI as one of the companion note's three indexes. The r^*-gap FCI always uses the Laubach–Williams r^*. The tables change only how the two-year yield is adjusted:

- no adjustment (nominal);
- minus expected inflation and the Laubach–Williams r^* (baseline);
- minus expected inflation and Kiley (2020)'s estimate of r^*;
- minus a market-based measure of the neutral rate: the one-year rate nine years ahead (the 9–10-year forward rate), from the Federal Reserve Board's estimated yield curve (Gürkaynak et al., 2007). This rate is nominal, so it replaces both the expected-inflation and r^* adjustments;
- minus the average of the two-year yield over the past 8 years.

Three results stand out.

First, financial conditions are essential for unemployment. Without the FCI, the unemployment response is weak with a model-based r^*. It is significantly negative with the nominal rate, the forward rate or the 8-year average. With any of the FCIs, it is positive at every horizon and significant at the 90 percent level in almost all cases.

Second, the price effect is strong whenever the FCI is measured relative to r^*. With the r^*-gap FCI, the price level falls 0.3 to 0.7 percent after 24 months. This holds for every adjustment of the two-year yield, and even with no adjustment. This fall is significant at the 90 percent level for every lag length. With the nominal or real FCI, the price effect is strong when the two-year yield is adjusted with a model-based r^*. It is weaker with the forward rate or the 8-year average. The forward rate also includes a term premium (the extra yield investors require to hold long-term bonds), which fell over the sample.

Third, Kiley's r^* gives results close to those with Laubach–Williams, though somewhat weaker for prices. The r^*-gap FCI always uses the Laubach–Williams r^*. So these results show that how the two-year yield is adjusted matters less once the FCI is measured relative to r^*. They do not show that the choice of r^* in the FCI is unimportant.

Table 2: Different adjustments of the two-year yield: unemployment gap (percentage points)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
Two-year yield: nominal									
none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
real	0.41**	0.32**	0.25†	0.69**	0.67**	0.57**	0.68**	0.74**	0.70**
r^*-gap	0.39**	0.31**	0.29**	0.58**	0.57**	0.55**	0.52**	0.57**	0.60**
Two-year yield: Laubach–Williams r^*-gap (baseline)									
none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
nominal	0.48**	0.43**	0.35**	0.63**	0.63**	0.58**	0.49**	0.53**	0.53**
real	0.46**	0.42**	0.36**	0.59**	0.59**	0.57**	0.45**	0.49**	0.51**
r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Two-year yield: Kiley r^*-gap									
none	0.13†	−0.02	−0.15†	0.26†	0.07	−0.11	0.30**	0.13	−0.03
nominal	0.44**	0.38**	0.32**	0.58**	0.60**	0.58**	0.43**	0.50**	0.56**
real	0.44**	0.38**	0.32**	0.55**	0.58**	0.57**	0.40**	0.48**	0.54**
r^*-gap	0.39**	0.36**	0.33**	0.45**	0.48**	0.52**	0.31**	0.39**	0.45**
Two-year yield: minus 9–10-year forward rate									
none	−0.20**	−0.20†	−0.23**	−0.25**	−0.17	−0.23†	−0.24**	−0.11	−0.14
nominal	0.26**	0.25**	0.18†	0.33**	0.42**	0.35**	0.26**	0.35**	0.35**
real	0.26**	0.26**	0.20†	0.31**	0.40**	0.35†	0.24**	0.33**	0.34†
r^*-gap	0.22**	0.23**	0.26**	0.26**	0.35**	0.40**	0.21**	0.30**	0.37**
Two-year yield: minus average over past 8 years									
none	−0.16†	−0.28**	−0.30**	−0.05	−0.18	−0.30†	0.02	−0.06	−0.18
nominal	0.30**	0.25**	0.21**	0.39**	0.43**	0.40**	0.27**	0.35**	0.38**
real	0.27**	0.21**	0.17†	0.36**	0.39**	0.36**	0.26**	0.33**	0.36**
r^*-gap	0.22**	0.17**	0.21**	0.33**	0.36**	0.41**	0.29**	0.37**	0.46**

See Table 1. The FCI is always one of the companion note's three indexes; the r^*-gap FCI uses the Laubach–Williams r^*. All specifications include the quadratic trend.

Table 3: Different adjustments of the two-year yield: core PCE price level (percent)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
Two-year yield: nominal									
none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
real	−0.09	−0.11	−0.13	−0.23	−0.34†	−0.33	−0.31	−0.51†	−0.50
r^*-gap	−0.20**	−0.22**	−0.21**	−0.32**	−0.50**	−0.52**	−0.32†	−0.61†	−0.72**
Two-year yield: Laubach–Williams r^*-gap (baseline)									
none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
nominal	−0.25**	−0.28**	−0.30**	−0.42**	−0.59**	−0.66**	−0.50**	−0.75**	−0.92**
real	−0.26**	−0.29**	−0.33**	−0.44**	−0.59**	−0.70**	−0.52**	−0.75**	−0.97**
r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Two-year yield: Kiley r^*-gap									
none	−0.23**	−0.16†	−0.18**	−0.37**	−0.34†	−0.29†	−0.44**	−0.48†	−0.41†
nominal	−0.23**	−0.24**	−0.21†	−0.39**	−0.50**	−0.41†	−0.43**	−0.61**	−0.53†
real	−0.25**	−0.25**	−0.25**	−0.40**	−0.51**	−0.46†	−0.44**	−0.62**	−0.58†
r^*-gap	−0.34**	−0.33**	−0.29**	−0.48**	−0.62**	−0.58**	−0.47**	−0.69**	−0.68**
Two-year yield: minus 9–10-year forward rate									
none	0.01	−0.04	−0.08	0.04	−0.04	−0.06	0.05	−0.06	−0.07
nominal	−0.05	−0.12	−0.14†	−0.07	−0.21	−0.25	−0.06	−0.22	−0.31
real	−0.06	−0.14†	−0.18†	−0.07	−0.21	−0.30†	−0.06	−0.21	−0.37
r^*-gap	−0.22**	−0.27**	−0.30**	−0.29**	−0.48**	−0.62**	−0.30†	−0.53**	−0.81**
Two-year yield: minus average over past 8 years									
none	−0.08	−0.06	−0.08	−0.25†	−0.27†	−0.23†	−0.37†	−0.47†	−0.45†
nominal	−0.13†	−0.15†	−0.16†	−0.27†	−0.38**	−0.39†	−0.31†	−0.49**	−0.56†
real	−0.14†	−0.15†	−0.18†	−0.28†	−0.37**	−0.42†	−0.32†	−0.49**	−0.61†
r^*-gap	−0.25**	−0.24**	−0.24**	−0.46**	−0.61**	−0.63**	−0.56**	−0.83**	−0.96**

See Table 2. The 8-year average and the Laubach–Williams trend in the two-year yield (two-year expected inflation plus r^*) are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic and recent years. The baseline drops March–December 2020 from the months used to fit the monthly equations. But 2020 values still enter the 2021 equations as lags. Table 4 therefore also drops 2021. It then ends the sample in December 2022, and separately in December 2019. In those two shorter samples, the daily model is also estimated only over the shorter period.

With the r^*-gap two-year yield and FCI, the unemployment response holds up. After 12 months, it is 0.26 to 0.31 percentage point when 2021 is dropped, 0.32 to 0.34 when the sample ends in 2022, and 0.42 to 0.44 when it ends in 2019. It is significant at the 90 percent level at almost every horizon and lag length. Without the FCI, its 90 percent interval includes zero in all three samples.

The price effect is unchanged when the sample ends in 2022, so the data since 2023 play no special role. It stays significant with 6 and 12 lags when 2021 is dropped (−0.43 and −0.63 percent after 24 months). But in the sample that ends in 2019, the price level barely moves. So the significant price effect comes from including 2020–22.

Figure 2 shows why these years are informative. After 2021, the r^*-gap two-year yield and the FCI rose by about 4 percentage points. This was the largest swing in the stance of policy relative to r^* in the sample, and core inflation then fell. Figure 3 drops one calendar year at a time. No single

year drives the results. The unemployment response after 12 months ranges from 0.26 (dropping 2021) to 0.41. The price level after 24 months ranges from −0.43 (dropping 2021) to −0.86. The 90 percent intervals exclude zero in every case. So the price result rests on the 2020–22 episode as a whole—a large tightening relative to r^* followed by falling inflation—not on any one year.

Table 4: Different samples: r^*-gap two-year yield, without and with the r^*-gap FCI

		12 months			24 months			36 months		
Sample	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Baseline (excl. Mar–Dec 2020)	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Also drop 2021	none	0.03	−0.03	−0.09	0.09	−0.01	−0.09	0.12	−0.01	−0.11
	r^*-gap	0.31**	0.26**	0.30**	0.38**	0.33**	0.38**	0.29**	0.28†	0.30**
End in 2022	none	0.11	0.07	−0.04	0.19†	0.11	−0.01	0.21†	0.13	0.01
	r^*-gap	0.32**	0.33**	0.34**	0.35**	0.38**	0.43**	0.23**	0.30**	0.37**
End in 2019	none	0.05	0.05	0.00	0.09	0.14	0.14	0.11	0.19	0.28
	r^*-gap	0.42**	0.42**	0.44**	0.45**	0.50**	0.60**	0.34**	0.42**	0.59**
B. Core PCE price level (percent)										
Baseline (excl. Mar–Dec 2020)	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Also drop 2021	none	−0.11†	−0.09	−0.13†	−0.18†	−0.25†	−0.35**	−0.22†	−0.38†	−0.59**
	r^*-gap	−0.16†	−0.21**	−0.28**	−0.24†	−0.43**	−0.63**	−0.29†	−0.60**	−0.93**
End in 2022	none	−0.25**	−0.20**	−0.21**	−0.42**	−0.49**	−0.51**	−0.51**	−0.70**	−0.83**
	r^*-gap	−0.34**	−0.33**	−0.30**	−0.51**	−0.68**	−0.71**	−0.56**	−0.87**	−1.02**
End in 2019	none	0.02	0.02	0.08†	0.06	0.02	0.03	0.10	0.02	−0.06
	r^*-gap	−0.02	−0.03	−0.05	0.02	−0.00	−0.12	0.03	−0.03	−0.22†

See Table 1. In the "End in" samples, the daily model is also estimated over the shorter period.

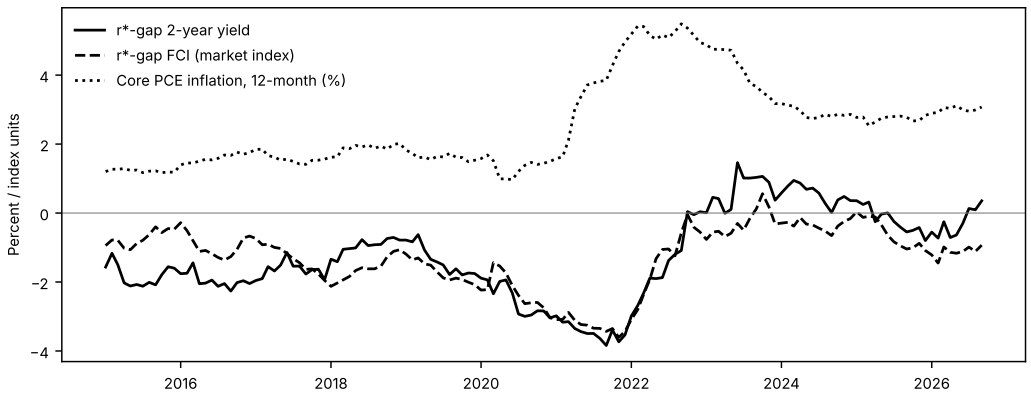


Figure 2: The r^*-gap two-year yield, the r^*-gap FCI and 12-month core PCE inflation, 2015–2026.

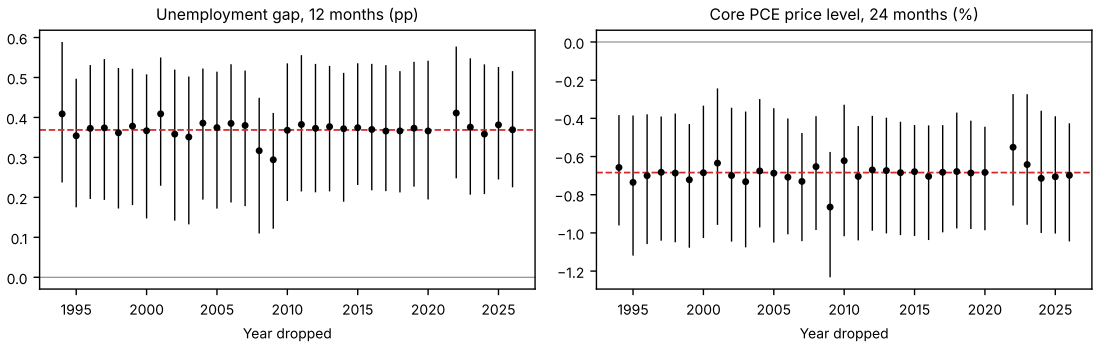


Figure 3: Dropping one year at a time. Baseline version (r^*-gap two-year yield and FCI, 6 lags; 100 bootstrap draws rather than 200, to limit computation). Each point drops one calendar year from the monthly equations. Bars: 90 percent intervals. Dashed line: full-sample estimate.

Industrial production. Appendix Table A1 adds industrial production to the monthly model. Without the FCI, industrial production rises after a tightening when the two-year yield is nominal or real. It does not move significantly when the two-year yield is measured relative to r^*. With the nominal or real FCI, it does not fall significantly. Only at the r^* step does it fall: by 0.84 to 1.05 percent after 12 months, significant at the 90 percent level for every lag length. Bauer and Swanson (2023a) find about 1.5 percent. In our model, industrial production recovers within about three years. In earlier exploratory work, we also added a commodity price index. Commodity prices then rose after a tightening. Our key assumption is least plausible for prices like these, which look ahead and respond strongly to demand.

Other surprise measures. Appendix Table A2 replaces the Nakamura–Steinsson surprise with three others and keeps everything else the same:

- the surprise with the part predicted by the Fed staff's forecasts for each FOMC meeting (the Greenbook) removed;
- the surprise of Bauer and Swanson (2023a), with the part predicted by public data removed (their posted series, which covers 201 of our announcements through 2019);
- the "pure policy" surprise of Jarociński and Karadi (2020), which keeps only announcements where the two-year yield and the S&P 500 move in opposite directions (157 announcements).

The choice of surprise series can matter for estimated effects (Bu et al., 2021; Brennan et al., 2025). In our design, a surprise enters only through its window moves, scaled to a 1 percentage point rise in the two-year yield, and through the timing of announcements. So this check really tests how much the FCI moves per point on the two-year yield. That ratio is 0.99 for the baseline, 1.07 for the Bauer–Swanson surprise and 1.09 for the Jarociński–Karadi surprise. The last ratio is higher partly by design: that surprise keeps announcements where stocks fall when yields rise, and stocks enter the FCI with a negative sign.

With these ratios, the effects are slightly larger than in the baseline. At the r^* step, unemployment rises 0.40 to 0.45 percentage point after 12 months, and the price level falls 0.55 to 0.77 percent after 24 months. The Greenbook-cleaned surprise gives results within 0.02 of the baseline. Without the FCI, all the surprises give almost identical effects. So our results do not depend on how the surprise is cleaned. Cleaning matters here only through the ratio of the FCI move to the two-year-yield move in the window.

A fuller measure of uncertainty. Appendix Table A4 uses 1,000 draws instead of 200.

In each draw, besides resampling the announcements, it re-estimates the daily model on randomly redrawn blocks of 66 consecutive days. It also rebuilds the monthly data from the estimated model, randomly flipping the signs of the errors, and re-estimates the monthly equations. The intervals barely change. At the r^* step, every ** and † marker is the same. Elsewhere, a few markers move between the 68 and 90 percent levels. The FCI weights and r^* are still held fixed.

Other checks. In earlier versions of the model, we added the gap between Baa corporate and Treasury yields to the daily model. This lowered the unemployment response by a third to a half. But no intraday credit data exist, so we could not measure how the credit spread moves in the announcement window. We do not pursue this here. Credit is the most important channel we leave out (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). We also tried replacing the daily model's path for the financial variables in the first one to two months with the path estimated directly from the data. That gives larger but much less precise effects.

5 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well. The hard part has been tracing their effects on the economy. We keep the surprise as it is and change only how its effects are traced. Two standard parts of the transmission mechanism explain much of the post-1994 puzzle. Policy works through financial conditions as a whole. And what matters for spending and inflation is the level of rates and financial conditions relative to a slow-moving neutral rate. When we trace surprises through a broad financial conditions index measured relative to r^*, the effects on unemployment, prices and industrial production have the standard signs. They are precisely estimated, the unemployment effects barely depend on the number of lags, and they are similar in size to the estimates of Bauer and Swanson (2023a).

The two parts play different roles. Financial conditions are essential for the unemployment effect, in every version and sample we try. Measuring financial conditions relative to r^* is essential for the price effect. When the FCI is measured relative to the Laubach–Williams r^*, the price effect is strong however the two-year yield is adjusted. The unemployment effect holds up however we treat the pandemic. The price effect depends mainly on the years since 2020, when the stance of policy relative to r^* moved most.

These findings complement the work on information effects and predictable surprises. Several mechanisms can operate at once. A natural next step is to trace cleaned or separated surprises through our approach. Our results have limits. They rest on the key assumption: policy reaches the economy through the financial variables in the model, and the economy responds to them the same way whatever moved them. They also rest on the time trend. They leave out credit spreads. And our intervals do not include uncertainty about r^* or the FCI weights. The broader lesson is that the fall in r^* over the past three decades is more than a fact about the level of interest rates. It also makes the effects of policy harder to measure, unless rates are measured relative to r^*.

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A Additional tables

Table A1: Responses with industrial production added to the monthly model

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	−0.04	−0.19†	0.26†	0.16	−0.04	0.26	0.30†	0.19
	included	0.41**	0.36**	0.22**	0.60**	0.59**	0.47**	0.51**	0.56**	0.49**
Real	none	0.12	0.03	−0.14†	0.24†	0.15	−0.02	0.24†	0.23	0.15
	included	0.41**	0.37**	0.24**	0.54**	0.53**	0.46**	0.43**	0.48**	0.45**
r^*-gap	none	0.16†	0.16†	0.03	0.30**	0.27†	0.14	0.33**	0.30†	0.25†
	included	0.44**	0.42**	0.34**	0.52**	0.50**	0.50**	0.42**	0.42**	0.42**
B. Industrial production (percent)										
Nominal	none	1.06**	1.26**	1.65**	1.03†	1.26†	1.64**	1.16†	1.00†	1.10†
	included	−0.16	−0.29	0.11	−0.09	−0.04	0.15	0.45	0.39	0.43
Real	none	0.64†	0.77**	1.13**	0.72	0.97†	1.28**	0.87†	0.85†	0.91†
	included	−0.33	−0.52	−0.23	−0.11	−0.11	−0.01	0.42	0.31	0.34
r^*-gap	none	0.13	−0.04	0.27	0.13	0.23	0.32	0.31	0.44	0.21
	included	−0.84**	−1.05**	−0.95**	−0.46†	−0.48	−0.80**	0.18	0.08	−0.21

See Table 1. Price-level responses are similar to Table 1. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8.

Table A2: Other surprise measures, with the FCI

		12 months			24 months			36 months		
Shock	Step	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nakamura–Steinsson	Nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Nakamura–Steinsson, Greenbook part removed	Nominal	0.43**	0.34**	0.27**	0.75**	0.73**	0.62**	0.75**	0.81**	0.76**
	r^*-gap	0.39**	0.36**	0.34**	0.46**	0.47**	0.48**	0.34**	0.39**	0.42**
Bauer–Swanson, predictable part removed	Nominal	0.47**	0.39**	0.32**	0.81**	0.79**	0.67**	0.81**	0.87**	0.80**
	r^*-gap	0.44**	0.41**	0.40**	0.52**	0.52**	0.54**	0.40**	0.44**	0.46**
Jarociński–Karadi pure policy	Nominal	0.48**	0.40**	0.33**	0.82**	0.81**	0.69**	0.83**	0.89**	0.81**
	r^*-gap	0.45**	0.42**	0.41**	0.53**	0.54**	0.55**	0.41**	0.45**	0.47**
B. Core PCE price level (percent)										
Nakamura–Steinsson	Nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Nakamura–Steinsson, Greenbook part removed	Nominal	−0.08	−0.11	−0.11	−0.24	−0.35†	−0.30	−0.33	−0.54†	−0.48
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.52**	−0.67**	−0.71**	−0.56**	−0.83**	−0.95**
Bauer–Swanson, predictable part removed	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.32	−0.34	−0.55†	−0.51
	r^*-gap	−0.36**	−0.35**	−0.35**	−0.55**	−0.71**	−0.76**	−0.59**	−0.88**	−1.01**
Jarociński–Karadi pure policy	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.33	−0.34	−0.56†	−0.51
	r^*-gap	−0.36**	−0.36**	−0.35**	−0.55**	−0.72**	−0.77**	−0.60**	−0.89**	−1.03**

See Table 1. Each surprise is scaled to raise the two-year yield 1 percentage point in the window. Step: the nominal, or the r^*-gap, two-year yield and FCI.

Table A3: Main steps without the time trend

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	$-0.13^{\dagger}$	-0.27^{**}	-0.34^{**}	-0.01	$-0.19^{\dagger}$	-0.36^{**}	0.08	-0.08	$-0.25^{\dagger}$
	included	0.18^{**}	0.09	0.07	0.38^{**}	0.29^{**}	$0.19^{\dagger}$	0.42^{**}	0.37^{**}	0.24^{**}
Real	none	-0.02	$-0.14^{\dagger}$	-0.25^{**}	0.07	-0.10	-0.28^{**}	0.12	-0.03	$-0.21^{\dagger}$
	included	0.25^{**}	$0.18^{\dagger}$	0.12	0.40^{**}	0.33^{**}	0.24^{**}	0.38^{**}	0.36^{**}	0.26^{**}
r^*-gap	none	0.08	0.03	-0.08	$0.16^{\dagger}$	0.04	$-0.15^{\dagger}$	$0.20^{\dagger}$	0.05	-0.16
	included	0.30^{**}	0.32^{**}	0.34^{**}	0.38^{**}	0.39^{**}	0.41^{**}	0.34^{**}	0.38^{**}	0.37^{**}
B. Core PCE price level (percent)										
Nominal	none	0.01	0.06	0.01	-0.08	0.03	0.05	-0.17	-0.07	0.03
	included	-0.09	-0.07	-0.10	$-0.28^{\dagger}$	$-0.24^{\dagger}$	-0.23	$-0.42^{\dagger}$	$-0.41^{\dagger}$	$-0.36^{\dagger}$
Real	none	$-0.10^{\dagger}$	-0.02	-0.07	$-0.19^{\dagger}$	-0.08	-0.06	$-0.26^{\dagger}$	-0.16	-0.08
	included	$-0.17^{\dagger}$	$-0.14^{\dagger}$	-0.18^{**}	$-0.33^{\dagger}$	$-0.32^{\dagger}$	$-0.31^{\dagger}$	$-0.44^{\dagger}$	-0.47^{**}	$-0.41^{\dagger}$
r^*-gap	none	$-0.15^{\dagger}$	$-0.13^{\dagger}$	-0.19^{**}	$-0.22^{\dagger}$	$-0.26^{\dagger}$	$-0.31^{\dagger}$	$-0.28^{\dagger}$	-0.39	$-0.44^{\dagger}$
	included	$-0.18^{\dagger}$	-0.25^{**}	-0.33^{**}	$-0.28^{\dagger}$	-0.48^{**}	-0.61^{**}	$-0.37^{\dagger}$	-0.68^{**}	-0.84^{**}

As Table 1, but without the quadratic time trend.

Table A4: Main steps with a fuller measure of uncertainty

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	$-0.13^{\dagger}$	-0.27^{**}	$0.36^{\dagger}$	0.10	$-0.19^{\dagger}$	0.53^{**}	0.29	-0.00
	included	0.44^{**}	0.36^{**}	0.28^{**}	0.76^{**}	0.75^{**}	0.63^{**}	0.76^{**}	0.83^{**}	0.77^{**}
Real	none	$0.14^{\dagger}$	-0.04	-0.20^{**}	0.33^{**}	0.12	$-0.13^{\dagger}$	0.42^{**}	0.23	-0.01
	included	0.44^{**}	0.38^{**}	0.29^{**}	0.65^{**}	0.65^{**}	0.59^{**}	0.57^{**}	0.64^{**}	0.64^{**}
r^*-gap	none	0.19^{**}	0.10	-0.03	0.34^{**}	$0.21^{\dagger}$	0.04	0.38^{**}	$0.25^{\dagger}$	0.07
	included	0.40^{**}	0.37^{**}	0.36^{**}	0.47^{**}	0.48^{**}	0.50^{**}	0.35^{**}	0.40^{**}	0.43^{**}
B. Core PCE price level (percent)										
Nominal	none	-0.03	-0.01	-0.02	-0.16	-0.16	-0.07	-0.26	-0.34	-0.18
	included	-0.09	-0.11	-0.11	-0.24	$-0.36^{\dagger}$	$-0.31^{\dagger}$	-0.33	$-0.54^{\dagger}$	$-0.49^{\dagger}$
Real	none	$-0.14^{\dagger}$	-0.08	-0.10	$-0.25^{\dagger}$	$-0.23^{\dagger}$	-0.16	$-0.33^{\dagger}$	$-0.37^{\dagger}$	-0.26
	included	$-0.17^{\dagger}$	$-0.17^{\dagger}$	$-0.18^{\dagger}$	$-0.30^{\dagger}$	$-0.40^{\dagger}$	$-0.36^{\dagger}$	$-0.35^{\dagger}$	$-0.54^{\dagger}$	$-0.49^{\dagger}$
r^*-gap	none	-0.27^{**}	-0.22^{**}	-0.23^{**}	-0.45^{**}	-0.51^{**}	-0.52^{**}	-0.55^{**}	-0.70^{**}	-0.80^{**}
	included	-0.34^{**}	-0.33^{**}	-0.33^{**}	-0.53^{**}	-0.68^{**}	-0.72^{**}	-0.57^{**}	-0.85^{**}	-0.97^{**}

As Table 1, with 1,000 draws. Each draw resamples the announcements, re-estimates the daily model on resampled 66-day blocks, and rebuilds and re-estimates the monthly data with randomly sign-flipped errors. r^* and the FCI weights are held fixed.